

Theorem proving

Direct proof · contrapositive · contradiction · disproof by counterexample

Today's goal: turn a mathematical statement into a clear argument that shows **why it must be true in every case it claims**.

A proof is not a list of examples. It is a chain of valid deductions from stated assumptions, definitions, and previously known facts.

Session	Date
5 of 15	Fri 11 Sep 2026

Where session 5 sits

1 · Course overview and school-maths refresher

2 · Sets and set operations

3 · Functions

4 · Function families

5 · Theorem proving

6 · Mathematical induction

7 · Elementary combinatorics and binomial theorem

8 · Midterm

9 · GCD and divisibility

10 · Modular arithmetic

11 · Vectors

12 · Polynomials

13 · Complex numbers

14 · Course recap and exam preparation

15 · Final exam

TODAY Direct proof, contrapositive, contradiction, and counterexamples — the tools used to justify the precise claims introduced in earlier sessions.

Function families, in four ideas

TRANSFORMATIONS · SINE

Inside moves the input;
outside moves the output. In
 $A \sin(B(t - h)) + D$ with $B > 0$: amplitude $|A|$,
period $2\pi/B$, midline $y = D$.

POLYNOMIALS

The leading term decides the ends;
roots give factors. $p(r) = 0 \iff (x - r)$ is a
factor, so degree n means at most n real roots.

EXPONENTIALS & LOGS

A log is an exponent:
 $\log_b x = y \iff b^y = x$, so its input must be
positive. $\log(uv) = \log u + \log v$ **only if** $u, v > 0$;
 $\log(u + v)$ does not split in general.

GROWTH

"Eventually" means from some point
on. $\log_2 n < n^2 < 2^n$ for every $n \geq 5$, not for every
 n : at $n = 2$, $n^2 = 2^n$.

WARM-UP

One pair disproves the $\log_2(u + v)$ rule. Why can't examples prove the $\log_2(uv)$ law?

A theorem is more than a pattern

DEFINITION A **mathematical statement** is a declarative sentence with a definite truth value: once all variables, domains, and quantifiers are fixed, it is either true or false.

DEFINITION A **theorem** is a mathematical statement established by a valid proof from accepted definitions, axioms, and earlier results.

SENTENCE	STATEMENT?	WHY?
" $x + 1 = 4$ "	No	It is an open sentence : its truth depends on the unspecified value of x .
"For every real x , $x^2 \geq 0$ "	Yes	It makes one claim about all real numbers.
"There is an integer n with $n^2 = 9$ "	Yes	It says that at least one integer works.

QUANTIFIERS "For every" means **all allowed values**; "there is" means **at least one allowed value**.

An implication has a hypothesis and a conclusion

DEFINITION An implication has the form

$$P \implies Q,$$

read as "if P , then Q ."

- P is the **hypothesis** (or condition).
- Q is the **conclusion**.

EXAMPLE · WITH ITS DOMAIN For every $n \in \mathbb{Z}$, if n is a multiple of 4, then n is even.

THE WORDS, IN ALGEBRA $\mathbb{Z}$ is the set of integers. "Multiple of 4" means $n = 4k$ for some $k \in \mathbb{Z}$; "even" means $n = 2m$ for some $m \in \mathbb{Z}$.

ONE DIRECTION ONLY An implication only tells us what follows **when its hypothesis holds**. It does not say that every even integer is a multiple of 4.

Do not confuse an implication with its converse

NOTATION Fix one domain and consider an implication $P \Rightarrow Q$ in that domain. Here, $\neg P$ means "not P ."

NAME	FORM
original statement	$P \Rightarrow Q$
converse	$Q \Rightarrow P$
contrapositive	$\neg Q \Rightarrow \neg P$

THEOREM The original statement and its contrapositive are **logically equivalent**: over the same domain, they are always both true or both false. The converse is a different statement and may have a different truth value.

EXAMPLE "multiple of 4 $\Rightarrow$ even" is true. Its converse, "even $\Rightarrow$ multiple of 4," is false: 2 is even and is not a multiple of 4.

Read the logical direction

ACTIVITY 1 PAIRS

10 MIN

Let $P(n)$ mean " n is even" and $Q(n)$ mean " n^2 is even," where $n \in \mathbb{Z}$.

1. Write the converse of $P(n) \Rightarrow Q(n)$ in words.
2. Write the contrapositive of $P(n) \Rightarrow Q(n)$ in words.
3. For "if n is a multiple of 4, then n is even," test its converse over $\mathbb{Z}$. Is it true or false? Justify your answer.

Converse: swap. Contrapositive: swap and negate.

METHOD For $P \Rightarrow Q$:

1. **Converse**: swap the two parts: $Q \Rightarrow P$.
2. **Contrapositive**: swap the parts **and** negate both: $\neg Q \Rightarrow \neg P$.

PARITY OF INTEGERS For $t \in \mathbb{Z}$, " t is odd" means $t = 2k + 1$ for some $k \in \mathbb{Z}$. Every integer is either even or odd, not both. So "not even" means "odd," and "not odd" means "even."

REQUESTED STATEMENT

ANSWER

converse of "even $\Rightarrow$ square even"

"if n^2 is even, then n is even"

contrapositive

"if n^2 is odd, then n is odd"

converse of "multiple of 4 $\Rightarrow$ even"

false; $n = 2$ is a counterexample

KEY POINT Proving a converse proves a different statement.

To prove $P \Rightarrow Q$, prove $P \Rightarrow Q$ itself or prove its equivalent contrapositive $\neg Q \Rightarrow \neg P$.

Direct proof: start from the hypothesis

THE SHAPE To prove a universal implication

for every $x \in D$, $P(x) \Rightarrow Q(x)$,
use this reliable structure.

NEW NUMBERS The hypothesis may provide new numbers: " a is even" supplies an integer r with $a = 2r$. That r is not a conveniently chosen value.

METHOD

1. Let $x \in D$ be **arbitrary**, and assume $P(x)$.
2. Replace words in the hypothesis by their definitions.
3. Use algebra or facts already known to be true.
4. Show $Q(x)$. Because x was arbitrary, the claim holds for every $x \in D$.

NEVER Never assume $Q(x)$ at the start: that is the conclusion you are trying to prove.

The sum of two even integers is even

CLAIM For every $a, b \in \mathbb{Z}$, if a and b are even, then $a + b$ is even.

PROOF Let $a, b \in \mathbb{Z}$ be arbitrary and assume that both are even. By the definition of even, there are integers r and s such that

$$a = 2r \quad \text{and} \quad b = 2s.$$

Therefore

$$a + b = 2r + 2s = 2(r + s).$$

Since $r + s \in \mathbb{Z}$, the number $a + b$ is 2 times an integer. This is exactly the definition of " $a + b$ is even." ■

Write a short direct proof

ACTIVITY 2 PAIRS

12 MIN

Prove the following statement directly:

If a is an even integer and b is an integer, then ab is even.

Use this checklist:

1. Let $a, b \in \mathbb{Z}$ be arbitrary and assume that a is even.
2. Use the definition: write $a = 2k$ for some $k \in \mathbb{Z}$.
3. Rewrite ab in the form $2 \times (\text{an integer})$.

A direct-proof walkthrough

METHOD

1. Open with arbitrary allowed values and the hypothesis, not the conclusion.
2. Substitute the definition into the expression you need to study.
3. Finish by showing that your answer has the form required by the conclusion.

PROOF Let $a, b \in \mathbb{Z}$ be arbitrary, and assume a is even.

Since a is even, $a = 2k$ for some $k \in \mathbb{Z}$. Thus

$$ab = (2k)b = 2(kb).$$

Because $k, b \in \mathbb{Z}$, we have $kb \in \mathbb{Z}$. Therefore ab is 2 times an integer, so ab is even. ■

THE TRAP $2(kb)$ only proves "even" after we state that kb is an integer.

When a direct proof is not the clearest route

Sometimes it is hard to work forwards from the hypothesis to the conclusion. For a claim $P \Rightarrow Q$, compare the starting points:

- **direct proof**: assume P , derive Q ;
- **contrapositive**: assume $\neg Q$, derive $\neg P$;
- **contradiction**: assume P and $\neg Q$, derive something impossible.

The last route works because the negation of $P \Rightarrow Q$ is " P is true **and** Q is false."



Contrapositive: prove the equivalent statement

EQUIVALENCE For each x in a fixed domain D , the statements

$$P(x) \Rightarrow Q(x) \quad \text{and} \quad \neg Q(x) \Rightarrow \neg P(x)$$

say the same thing in a different form: they are both true or both false.

The original fails exactly when $P(x)$ is true and $Q(x)$ is false. The contrapositive fails in exactly that same situation.

METHOD To prove a universal implication by contraposition:

1. Let $x \in D$ be arbitrary and assume $\neg Q(x)$.
2. Deduce $\neg P(x)$.
3. Because x was arbitrary, conclude $P(x) \Rightarrow Q(x)$ for every $x \in D$.

WHEN TO USE IT When "not Q " is easier to work with than P . You prove the contrapositive, not the converse.

If n^2 is even, then n is even

CLAIM For every $n \in \mathbb{Z}$, if n^2 is even, then n is even.

CHANGE THE VIEWPOINT Instead of starting from " n^2 is even," prove the contrapositive:

If n is odd, then n^2 is odd.

For integers, "not even" means "odd," and "not odd" means "even."

PROOF Let n be an arbitrary odd integer. By definition, $n = 2k + 1$ for some $k \in \mathbb{Z}$. Hence

$$\begin{aligned} n^2 &= (2k + 1)^2 \\ &= 4k^2 + 4k + 1 \\ &= 2(2k^2 + 2k) + 1. \end{aligned}$$

Because $k \in \mathbb{Z}$, we have $k^2 \in \mathbb{Z}$, and hence $j = 2k^2 + 2k \in \mathbb{Z}$. Thus $n^2 = 2j + 1$, so n^2 is odd. This proves the contrapositive; therefore the original claim is true. ■

Proof by contradiction: an impossible assumption

METHOD To prove a target statement S by contradiction:

1. Assume that S is false; that is, assume $\neg S$.
2. Reason carefully from that assumption.
3. Reach a contradiction: for example, both R and $\neg R$, or a fact known to be impossible.
4. Therefore the assumption $\neg S$ was false, so S is true.

CONDITIONAL TARGETS For a target $P \Rightarrow Q$, assuming its negation means assuming **both** P and $\neg Q$.

THE CONDITION The contradiction must be a consequence of the assumption, not merely a surprising result.

Set up $\sqrt{2}$

CLAIM $\sqrt{2}$ is irrational.

DEFINITION A real number is **rational** if it can be written as p/q , where $p, q \in \mathbb{Z}$ and $q \neq 0$. It is **irrational** if it is not rational.

Lowest terms: no positive integer greater than 1 divides both numerator and denominator. Any rational number can be written this way.

ASSUME, FOR CONTRADICTION Assume that $\sqrt{2}$ is rational.

1. Choose $p, q \in \mathbb{Z}$ with $q \neq 0$, no common positive divisor greater than 1, and $\sqrt{2} = \frac{p}{q}$.
2. Square both sides and use $(\sqrt{2})^2 = 2$:

$$2 = \frac{p^2}{q^2}, \quad \text{so} \quad 2q^2 = p^2.$$

3. Since $q^2 \in \mathbb{Z}$, this shows that p^2 is even. By the contrapositive theorem (n^2 even $\Rightarrow n$ even), p is even.

Finish $\sqrt{2}$

PROOF, CONTINUED

Since p is even, write $p = 2k$ for some $k \in \mathbb{Z}$. Substitute this into $2q^2 = p^2$:

$$2q^2 = (2k)^2 = 4k^2, \quad \text{so} \quad q^2 = 2(k^2).$$

Since $k^2 \in \mathbb{Z}$, the number q^2 is even. Because $q \in \mathbb{Z}$, the same theorem tells us that q is even as well.

Both p and q are divisible by 2. This contradicts the choice of a fraction in lowest terms. Therefore the assumption that $\sqrt{2}$ is rational is false, and $\sqrt{2}$ is irrational. ■

THE CONTRADICTION

"No common positive divisor greater than 1" conflicts with "both p and q are divisible by 2."

Choose a proof method before writing

ACTIVITY 3 GROUPS OF THREE

12 MIN

For each claim, choose the proof method that makes the starting condition easiest to use. Then write the first two lines, including the domain and the assumption.

CLAIM	YOUR METHOD
If $a, b \in \mathbb{Z}$ are odd, then $a + b$ is even.	?
For every $n \in \mathbb{Z}$, if n^2 is odd, then n is odd.	?
$\sqrt{2}$ is irrational.	?

Why these methods fit

CLAIM

GOOD WAY TO START

odd $a, b \in \mathbb{Z} \Rightarrow a + b$ even

Direct: let $a = 2r + 1$ and $b = 2s + 1$ for $r, s \in \mathbb{Z}$.

n^2 odd $\Rightarrow n$ odd, $n \in \mathbb{Z}$

Contrapositive: prove " n even $\Rightarrow n^2$ even." Let $n = 2k$ for $k \in \mathbb{Z}$; then $n^2 = 2(2k^2)$ and $2k^2 \in \mathbb{Z}$.

$\sqrt{2}$ irrational

Contradiction: assume it is rational and choose $\sqrt{2} = p/q$ in lowest terms.

METHOD

Choosing a method is part of writing a proof. Ask: which starting assumption gives me algebra I can use straight away, and what definition will my final line need?

A counterexample can disprove a "for every" claim

NEGATION RULES

$\neg(\text{for every } x \in D, P(x)) \iff \text{there exists } x \in D \text{ with } P(x) \text{ false.}$

$\neg(\text{there exists } x \in D \text{ with } P(x)) \iff \text{for every } x \in D, P(x) \text{ is false.}$

COUNTEREXAMPLE

To disprove "for every $x \in D, P(x)$," give a value a with (1) $a \in D$ and (2) $P(a)$ false. Here, D is the set of allowed values.

EXAMPLES

FALSE "FOR EVERY" STATEMENT

COUNTEREXAMPLE

Every prime number is odd.

2 is prime and even.

For every real x , $x^2 \geq x$.

At $x = \frac{1}{2}$, $x^2 = \frac{1}{4} < \frac{1}{2} = x$.

BOTH DIRECTIONS

One allowed $a \in D$ with $P(a)$ true proves "there exists $x \in D$ with $P(x)$," but normally does **not** prove a claim about every value. To disprove "there exists," show that no allowed value works.

Disprove, prove, or explain why one example is not enough

ACTIVITY 4 PAIRS

12 MIN

For each statement, decide whether it is true or false. Give the requested justification. When using a counterexample, name the value, confirm that it is in the domain, and show the failed equality or inequality.

1. For every real x , $x^2 \geq 0$. If true, state the known fact you use.
2. For every integer n , $n^2 + n$ is odd. If false, give a counterexample.
3. For every real x , $(x + 1)^2 = x^2 + 1$. If false, give a counterexample.
4. There exists a real x with $x^2 = -1$. Explain why checking a few values is not a disproof; use a fact about **every** real x instead.

Check "every" or "there exists" first

ITEM	REASONING
1	True. A square of a real number is non-negative: $x^2 \geq 0$ for every $x \in \mathbb{R}$.
2	False. The allowed integer $n = 1$ gives $n^2 + n = 2$, which is even, not odd.
3	False. The allowed real number $x = 1$ gives left side 4 and right side 2.
4	False. For every $x \in \mathbb{R}$, $x^2 \geq 0 > -1$, so no real x can satisfy $x^2 = -1$.

METHOD For item 4, a single failed trial is not enough: the claim says **there exists** a value. We disprove it by showing that **every** real value fails.

Four common proof mistakes

RECALL For the same domain, **not (every value has the property)** means **there exists a failure**; **not (there exists a value with the property)** means **every value fails**.

PITFALL	FIX
Proving $Q \Rightarrow P$ when the question asks for $P \Rightarrow Q$.	Label the hypothesis and conclusion before you start.
Assuming the conclusion.	Begin with P in a direct proof, with $\neg Q$ in a contrapositive proof, or with the negation of the whole target in a contradiction proof.
Negating "every" or "there exists" incorrectly.	Use the recall above: "for every" becomes "there exists" and vice versa, and the property itself is negated.
Giving a few examples as a proof for all values.	Examples test an idea; a proof must cover every case.

Work independently

TASK Complete questions 3 and 4. Then choose **one** full proof from questions 1 and 2.

1 Prove directly: if $m, n \in \mathbb{Z}$ are odd, then $m + n$ is even.

2 Prove by contraposition: for every $n \in \mathbb{Z}$, if n^2 is even, then n is even.

3 Disprove: for every real x , $x^2 + 1 > 2x$.

4 Write the **logical negation** of: "For every real x , $x^2 > x$."

RULE Write the method name and at least one sentence explaining why your proof or counterexample works. For a counterexample, state why the chosen value belongs to the stated domain.

What today was about

- State the **domain** and quantifier first. An implication has a **hypothesis** and a **conclusion**.
- Its converse is a different statement; over the same domain, its contrapositive is true in exactly the same cases.
- A **direct proof** starts from the hypothesis; a **contrapositive proof** starts from "not the conclusion"; a **contradiction proof** starts from the negation of the whole target.
- One valid **counterexample** — an allowed value where the property fails — disproves a statement that claims something is true for every allowed value.
- Remember: "not every" means "there exists a failure," and "not 'there exists'" means "every value fails."

Homework 5

Eight statements to prove or disprove.
Name the method at the top of each solution and explain each key step.

Next session: **mathematical induction** — a method for proving a statement for infinitely many integers.

implication

converse

contrapositive

negation

quantifier

contradiction

counterexample